Mercury-201

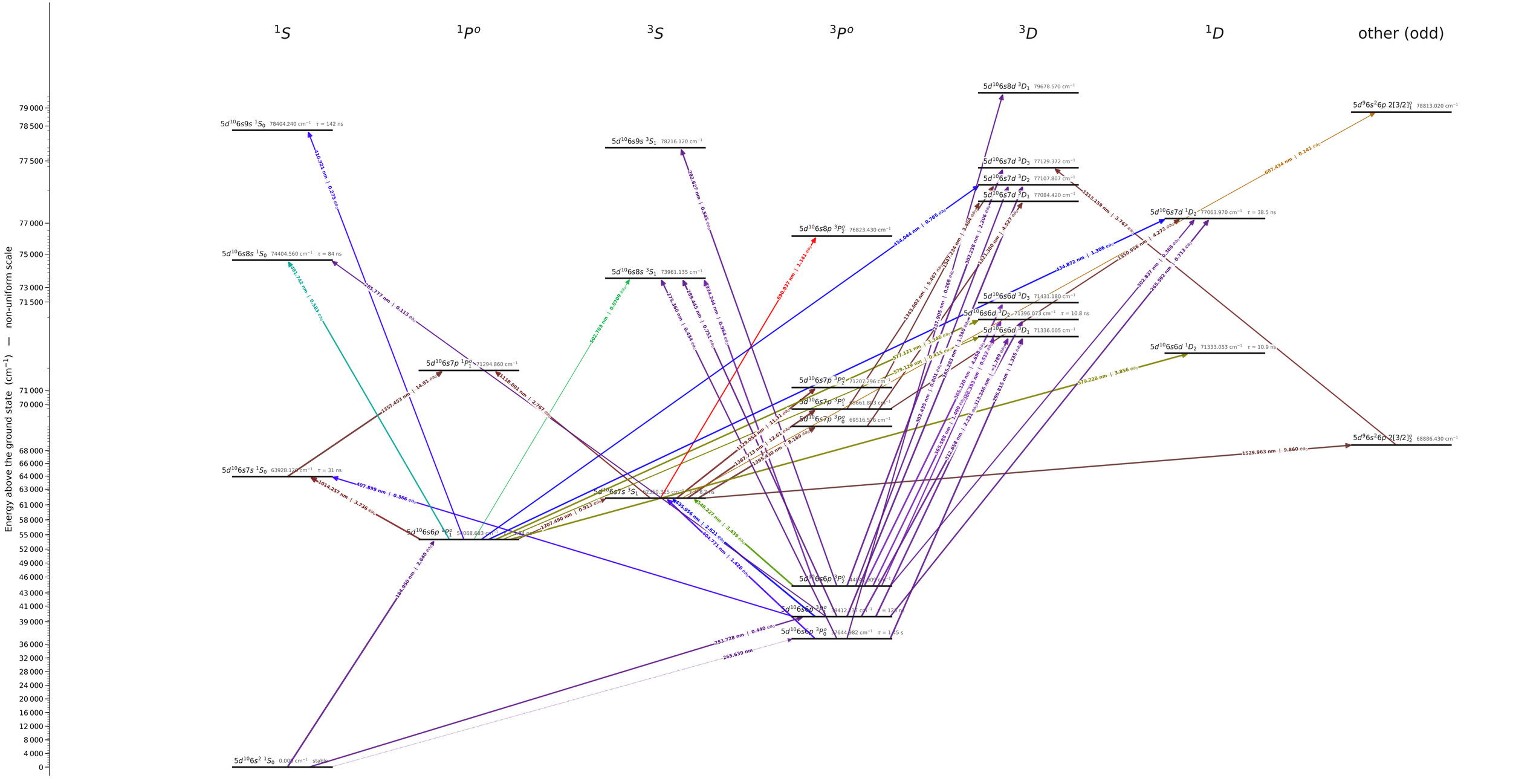
Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000

^{201}Hg (Hg I), nuclear spin I = 3/2. Strongest classified transitions below 2 µm (NIST): 27 levels, 47 lines.

Hg^+ ionisation limit 84184.15 cm^{-1} = 10.43750 eV (118.787 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\|\langle J\|er\|J'\rangle\|$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^1P^o_1$	184.9499	–	1620.9383	2.640	7.440e+08	118.8	99.7	184.9 nm resonance line
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_1$	253.7283	253.6521	1181.5491	0.440	7.992e+06	1.272	100	253.7 nm intercombination line: MOT / cooling, optical pu…
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_0$	265.6393	265.5603	1128.569561	–	6.920e-01	1.101e-07	100	265.6 nm optical lattice clock transition
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s8s\ ^1S_0$	285.7770	285.6930	1049.0434	0.113	1.100e+06	1.895	9.24	285.7 nm
$5d^{10}6s6p\ ^3P^o_0 - 5d^{10}6s6d\ ^3D_1$	296.8150	296.7283	1010.0315	1.335	4.600e+07	–	–	296.7 nm
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_2$	312.6579	312.5673	958.8513	2.231	6.600e+07	14.74	71.3	312.7 nm
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_1$	313.2462	313.1555	957.0505	≈1.789	7.036e+07	–	–	313.2 nm, optical pumping out of the 253.7 nm MOT
$5d^{10}6s6p\ ^3P^o_2 - 5d^{10}6s6d\ ^3D_3$	365.1198	365.0158	821.0797	4.658	1.290e+08	–	–	365.0 nm
$5d^{10}6s6p\ ^3P^o_0 - 5d^{10}6s7s\ ^3S_1$	404.7707	404.6564	740.6476	1.426	2.070e+07	19.41	17	404.7 nm
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s7s\ ^1S_0$	407.8988	407.7837	734.9677	0.366	4.000e+06	5.134	12.4	407.8 nm
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s7s\ ^3S_1$	435.9561	435.8336	687.6666	2.621	5.600e+07	19.41	45.9	435.8 nm
$5d^{10}6s6p\ ^1P^o_1 - 5d^{10}6s8s\ ^1S_0$	491.7418	491.6045	609.6543	0.583	5.800e+06	1.895	48.7	491.6 nm
$5d^{10}6s6p\ ^3P^o_2 - 5d^{10}6s7s\ ^3S_1$	546.2267	546.0750	548.8425	3.439	4.900e+07	19.41	40.2	546.1 nm
$5d^{10}6s6p\ ^1P^o_1 - 5d^{10}6s7s\ ^1S_0$	1014.2567	1013.9787	295.5785	3.736	2.710e+07	5.134	84	1014 nm
$5d^{10}6s6p\ ^1P^o_1 - 5d^{10}6s7s\ ^3S_1$	1207.4900	1207.1596	248.2774	0.913	3.200e+05	19.41	0.262	1207 nm intercombination branch of 7s 3S1

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{201}Hg hyperfine structure (I = 3/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$5d^{10}6s6p\ ^3P^o_1$	-5454.569(3)	–	F=1/2: +13636.42 F=3/2: +5454.57 F=5/2: -8181.85
$5d^{10}6s6p\ ^3P^o_2$	-3352.0292(8)	–	F=1/2: +15084.13 F=3/2: +10056.09 F=5/2: +1676.02 F=7/2: -10056.09
$5d^{10}6s6p\ ^1P^o_1$	1316.000	–	F=1/2: -3290.00 F=3/2: -1316.00 F=5/2: +1974.00

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_1$	253.728	196-198	4179.95(3)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_1$	253.728	200-198	-4805.52(4)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_1$	253.728	202-198	-10101.224(22)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_1$	253.728	204-198	-15311.60(4)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_1$	253.728	199-198	-652(6)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_1$	253.728	201-198	-6410(6)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_0$	265.639	199-201	5729.668(8)
$5d^{10}6s^2\ ^1S_0 - 5d^{10}6s6p\ ^3P^o_0$	265.639	198-199	654.48
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_2$	312.658	196-198	-289.4(12)
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_2$	312.658	200-198	328.0(6)
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_2$	312.658	202-198	686.7(6)
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_2$	312.658	204-198	1038.3(6)
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_1$	313.246	196-198	-307.5(10)
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_1$	313.246	200-198	347.2(6)
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_1$	313.246	202-198	725.5(6)
$5d^{10}6s6p\ ^3P^o_1 - 5d^{10}6s6d\ ^3D_1$	313.246	204-198	1097.8(6)

Shift = ν(first isotope) − ν(second isotope).